

Investigation of a Tolerable Time Delay in SEMG-based Elbow Assistive Device

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Abstract: Surface electromyography (SEMG) signal has been widely used to detect the intention of the user in assistive device, because the onset time of the SEMG signal is observed before the muscle force development. The time between the onset of SEMG activity and the development of force is defined as electromechanical delay (EMD). Owing to the existence of EMD, an SEMG-driven approach has advantage of early detection of the intention compared to other approaches, especially to a force sensor-based approach. However, few studies have been done on the investigation of the timing of the assistive force considering EMD. This study suggests the proper timing of the assistive force for better performance by analyzing the onset of muscle force development. Simultaneous measurements of four channels of SEMG, interaction torque, control input corresponding to the assistive torque, and joint angle allowed the investigation of the quantitative relation between the onset of SEMG and timing of the assistive torque. The timing of the assistive force was manipulated by varying the amount of artificial time delay. The experiments were carried out for 3 subjects, who were informed to complete 4 repetitive elbow flexion/extension motions as quickly as possible, with the help of the assistive device. Task completion time was examined as a measure of the performance of the assistance, and cross-correlation analysis was conducted. We observed that the reaching time and level of SEMG activation were reduced when the delay was consistent with the EMD of the user.

Keywords: SEMG, assistive device, exoskeleton, electromechanical delay

1. INTRODUCTION

Surface electromyography (SEMG) has been used as the control command for human assistive devices. One of the primary considerations in assistive devices is the detection of user intention, and SEMG is known for its merit to allow early identification of user's intention. Other than SEMG, mechanical signals such as joint torque or joint angle are also traditionally used, but they occur after or simultaneously with the actual muscle force activation, therefore enabling one to figure out user's intention only after the muscle activity. This characteristic of SEMG is quantitatively defined as electromechanical delay (EMD), which refers to the time lag between the onset of SEMG and the onset of force activation. EMD is known to be caused by an action potential propagation on muscle membrane, the coupling processes of excitation-contraction, and the stretching of the series elastic component, and shows difference depending on the person and on muscle [1], [2]

Although it has been generally agreed that this spare time owing to EMD can be one of the advantages of SEMG-driven control, there have not been enough researches validating the amount of advantages or discussing how to make good use of the early recognition of user intention. Instead, studies about SEMG in assistive devices have been mainly focused on how to interpret user's intention accurately. The techniques are usually based on pattern recognition algorithms [3], [4], [5], [6] or muscular-to-mechanical model such as Hill-based musculoskeletal model or proportional model [7], [8], [9], [10]. These algorithms correspond to the signal processing block as shown in Fig. 1, responsible

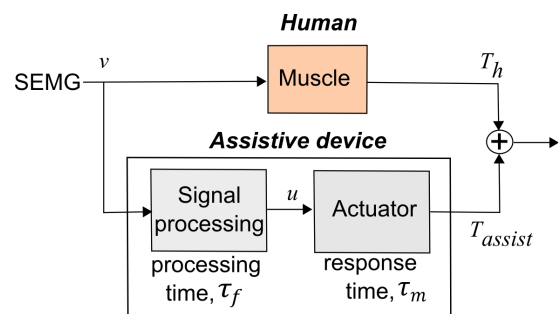


Fig. 1 Block diagram of general SEMG-driven assistive device.

for transforming raw SEMG to intended movement and torque activation level. In addition to modeling the signal processing block, which many previous researches have already spent their most effort on, deeper understanding of temporal delay is also required. In Fig. 1, time lapse between EMG stimulation and human torque (T_h) emergence is defined as EMD. Likewise, there also exist essential delay terms inside the assistive system. Because of the stochastic property of SEMG signal, the raw SEMG signals should be always smoothed before using as control input, causing an unavoidable phase delay to the original signal. The amount of this processing delay depends on parameters of smoothing methods; window size of mean absolute values (MAV) or cut-off frequency of a low-pass filter. Therefore, the operating timing of assist force (T_{assist}) depends on EMD, processing delay and response time of actuator of the device.

Therefore, in general, two torque signals (T_h , T_{assist}) are not generated simultaneously and the problem is that the existence of large delay between the intended torque from human and actuation of assist torque could

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bring fatal abnormality to the system. According to Ionescu[11], delayed feedback control might disturb the intended movement of human, or intimidate comfort and safety of user. In spite of the existence of inevitable delay terms that can crucially affect the performance of the system, majority of the existing SEMG-driven devices do not concern how the time delay should be handled. Most of them apply the processed SEMG signal instantly to the actuator, without taking timing issues into account. One might claim that the processing delay can be cancelled with the help of EMD, making no effective delay. However, it should be noted that early actuation timing has never been guaranteed, either. There exists a strong demand for the quantitative analysis to find the proper timing of the actuation, or to suggest its tolerable range. In this paper, the tolerable time delay between the SEMG onset and the actuation of the assistive torque is investigated by evaluating the performance of the assistive device under different conditions of the artificial delay. The robotic assistive device was designed to supply assistance to the elbow flexion/extension motion, and was controlled by only the users' SEMG signals. Artificial time delay τ_d was added algorithmically before the motor actuation, and task completion time was examined to evaluate the device performance. In order to estimate the delay between SEMG and torque signal, cross-correlation analysis between two signals were conducted, as in the Stokes' and Ionescu's studies[7], [11]. Through this investigation, the effect of time delay will be analyzed, and a new criterion will be suggested for the tolerable range of time delay between the SEMG activation and assistive force actuation. The rest of this paper is organized as follows: Section II reports the methods including signal processing, experimental setup and protocol, Section III analyzes the experimental results, Section IV discusses about optimal delay using cross-correlation analysis, and Section V concludes the effect of the timing of assistive force based on the experimental results.

2. METHOD

For the purpose of figuring out the tolerable time delay between the onset of SEMG and the timing of assisting force, experiments with the elbow assistive device were carried out.

2.1. Placement of SEMG sensor

SEMG signals were recorded from four muscles of the right arm (Fig. 2); Biceps brachii, Brachioradialis, Anconeus, and Triceps brachii; the first two involved in elbow flexion and the other two involved in elbow extension. Anconeus, a small muscle for extension of elbow joint is selected for this study because it is recruited while low level of extension force is required.

2.2. Onset detection algorithm

A threshold-based algorithm is used to detect the onsets of SEMG and torque sensor. The detection function involves signal processing; rectification, low pass filtering, and transformation. The detector compares the func-

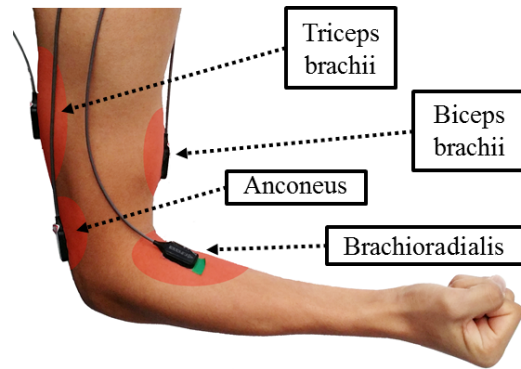


Fig. 2 Placements of the four SEMG electrodes.

tions with the fixed threshold values, and the detected onset is marked with a vertical red line, as shown in Fig. 3. It has advantages of intuitive detection and low computational effort. The SEMG signal is observed in advance of the torque signal, and this property is known as electro-mechanical delay (EMD), which is consistent with literature studies.

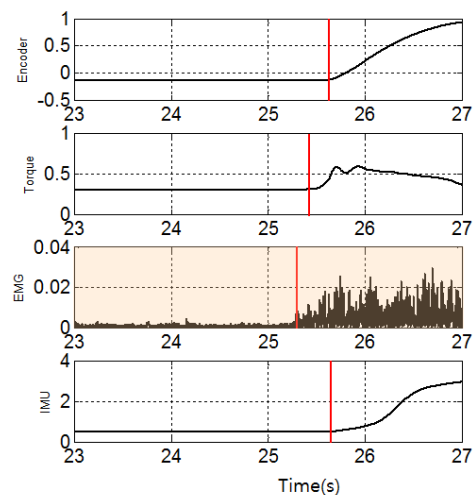


Fig. 3 Comparison of the detected onsets of SEMG activation, torque, elbow angle, and IMU using the threshold-based onset detection method.

2.3. Controller design

A SEMG-driven feed forward controller is designed to investigate the timing of the assistive torque. The total time gap between the onset of SEMG signal and the timing of the assistive torque consists of three components; an inevitable processing time during signal processing, an artificial time delay, and a motor response time, as shown in Fig. 4.

The processing time should be minimized in order to investigate the sole effect of the artificial delay. The other possibilities that could delay the actuation of assistive force were minimized with the utmost effort. In that context, only low pass filter (5 Hz cut off frequency) was used to smooth SEMG signal. The phase lag effect for

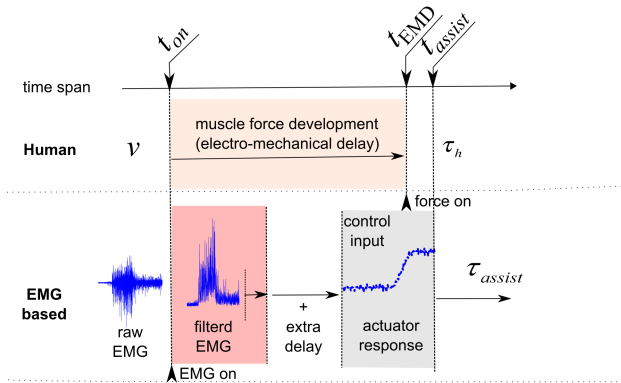


Fig. 4 The controller is designed to adjust the artificial delay.

the natural elbow flexion and extension motion, which has 0.4 ~ 0.5 Hz frequency, was about 25(ms). The total time gap was adjusted by adding an artificial delay in the control algorithm. From the detected onset of the SEMG, the eleven conditions of the artificial time delay was added, [1, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100 (ms)]. The motor response was examined using a step input and sinusoidal input response experiment and was identified as 12.5(ms). (Fig. 5) Because the SEMG-driven control is feed-forward type, the timing of the actuation is directly affected by the motor response. In summary, the inevitable delay in the assistive device was approximately 37.5(ms) from low pass filter (25(ms)) and motor response delay (12.5(ms)), and artificial delay from 1(ms) to 100(ms) was added.

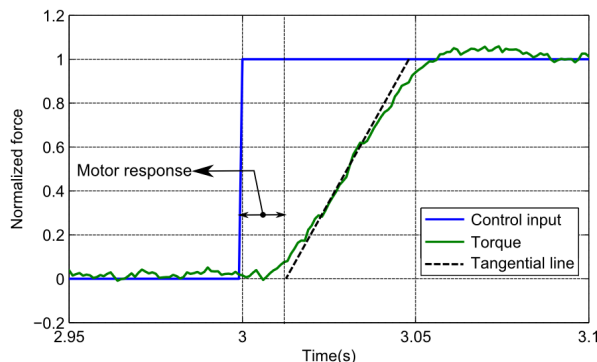


Fig. 5 Motor response was identified using step input analysis.

Based on the control algorithm, the vertical flexion and extension movement of right elbow was performed repeatedly with the assistance of the device. During the experiment, SEMG, interaction torque and joint angle data were acquired. The DC motor was controlled to assist the subject's movement, using only the SEMG signal. Three healthy subjects (20 - 30 years old) participated in the experiments.

2.4. Experimental setup

The customized testbed was built on an optical table which has adjustable height according to the height of the

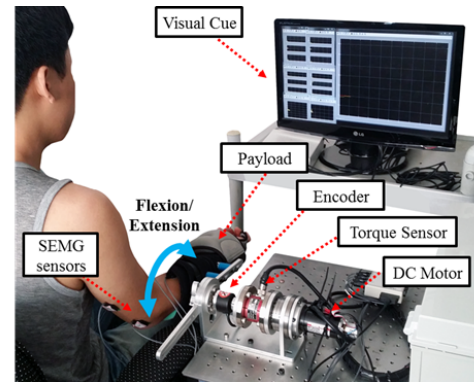


Fig. 6 Experimental setup.

subject's elbow joint, as shown in Fig. 6. The testbed, as an elbow assistive device, has a rigid link connected by a hinge joint and is installed by fixing its joint to the optical table. The lower arm brace is attached to the link which rotates actively, therefore the user can wear the device and rotates their lower arm to perform elbow joint motion with the device. Torque sensor (ST-100, CTA Plus, Korea), rotary encoder, and DC motor were implemented on the device joint, and were used to measure the interaction torque, joint angle, and to actuate the joint angular motion, respectively.

Bipolar noninvasive surface electrodes (Delsys) with built-in $\times 1000$ amplifiers are placed over the target muscles using medical adhesive tape. The electrodes have parallel bars (10(mm) length, 1(mm) width, 10(mm) inter-electrode distance), and are designed with built-in amplifiers (bandwidth: 20.450(Hz), Bandwidth Roll off: 80 (dB/decade)). A ground electrode was placed on the left ulna. The electrodes were connected to a data acquisition board (PCI 6221, National Instruments, TM, U.S.A.), and signals were sampled at 1000(Hz). The processing algorithm is implemented with the help of real-time windows target packet in Simulink (MATLAB, Mathworks). At the beginning of each running, the DC biases of EMG signals and the torque sensor were measured and removed. Therefore, subjects were informed to relax the muscles for the first three seconds of every trial.

2.5. Experimental protocol

At first, EMG electrodes were attached to the subject's arm. Then the subject wore the assistive device by tightly fastening his/her lower arm to the arm brace with Velcro, ensuring that the movements of the subject and the device, and their rotation axes correspond perfectly to each other. The height of the testbed and the position of the subject's body were adjusted so that the subject can conduct vertical elbow flexion/extension motion.

To normalize the user-dependency of the assisting torque, maximum voluntary contraction (MVC) was measured for each subject. The participants were to perform five arbitrary repetitions of flexion and extension while wearing the device but not being assisted, and the peak values from four SEMG channels during the whole

motion were recorded. Four maximum values are reflected to the motor gain value to decide how much the subject will be assisted by the device.

(1) Electromechanical delay (EMD) measurement

EMD of each muscle for every participant is essential information to analyze the effect of total time gap. To get the reliable value of EMD, five times of repeated activation and relaxation of the muscles were performed. Trapezoid-shaped cue was given to the participants with a period of seven seconds, to help them conduct the flexion/extension motion with consistent angular rate of $60^\circ/s$. The subject was aware of his/her own motion by monitoring the angular position of rotary encoder.

(2) Point-to-Point movement task

This experiment was designed to verify how the time gap between the SEMG and assistive force affects the performance of the task. The subject's task was to carry four or five reciprocating flexion/extension movements between the targeted angular positions. The target positions were defined as 0° and 60° . Subjects were informed to move their arm as quickly as possible over each targeted position, and their movement trials were only counted when the subject exceeded the targeted position. Before the task started, subjects were trained for the movement with the EMG-driven assistive device. A single set of the protocol consisted of the eleven different conditions of the artificial time delay [1, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100 (ms)], and five sets were conducted totally. In the first set, the eleven conditions were ordered in the ascending order, from 1 (ms) to 100 (ms). For the following sets, the sequence of conditions was determined according to the time performance of the previous set, in order to exclude the effect of motor adaptation. The condition that corresponds to the slowest performance in the past set was carried out first. The condition which achieved the fastest performance time was placed for the last condition of the next set.

3. EXPERIMENTAL RESULTS

3.1. EMD measurement result

The four SEMG signals and a torque sensor measurement were extracted to identify the EMD of the subject. The onset of each sensor was marked by red asterisk, as shown in Fig. 7. The EMD was identified by the difference between the onset of EMG and onset of torque sensor according to the definition of the EMD. The identified EMDs of three subjects were summarized in Table 1. The means of EMD were different among subjects and movement direction ranged from 47(ms) to 95(ms). The standard deviations of EMD were obtained up to 36(ms) due to the stochastic property of EMG signals.

3.2. Point-to-point movement result

The performance time of each condition was normalized by the mean value of the each set. The results commonly show that the minimum point of the mean of the performance time is occurred in the range between 10(ms)

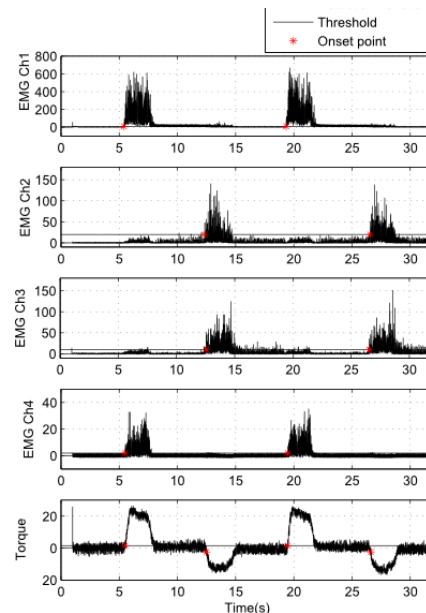


Fig. 7 Onset detections of EMG signals and torque sensor measurement. The threshold were chosen as three to twenty in accordance of the baseline noise level.

Table 1 Measured EMD by the onset detection method.

Subject Number	Age	Average EMD(standard deviation)	
		Flexion	Extension
S01	28	95(± 36)	79(± 22)
S02	23	93(± 27)	72(± 21)
S03	26	62(± 21)	47(± 25)

and 30(ms) of the artificial delay, as shown in Fig. 8; 30(ms) for subject 1, 20(ms) for subject 2, and 10(ms) for subject 3. In order to validate the difference of the

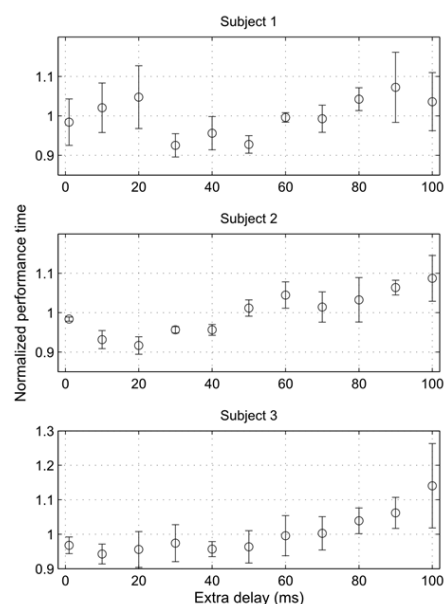


Fig. 8 Point-to-point task performance time varying with the artificial delay.

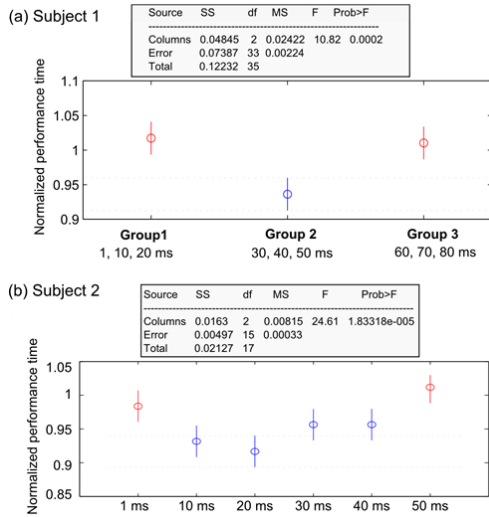


Fig. 9 (a) ANOVA test for subject 1 ($\alpha = 0.05$); Group 1 and Group 3 have means significantly different from Group 2, (b) ANOVA test for subject 2; The means of the normalized performance time in 20(ms) are significantly different those in 1(ms) and 50(ms)

means of each condition, ANOVA (analysis of the variance) test was carried with significance $\alpha = 0.05$. In case of subject 1, the conditions of (1, 10, 20 ms), (30, 40, 50 ms), and (60, 70, 80 ms) were defined as Group 1, Group 2, and Group 3, respectively. The means of Group 2 which involves a local minimum point 30(ms) was significantly different to the means of Group 1 and Group 3, as shown in Fig. 9 (p-value = 0.0002). In case of the subject 2, the means of the 20(ms) condition was significantly different with the means of 1(ms) and 50(ms) (p-value = 1.83318e-5)

4. DISCUSSION

In this study, we have focused on two important signals. The first one is DC motor input signal $V(t)$ which is represented as the linear combination of four SEMG signals and filtered by first order low-pass filter with 5(Hz) cut-off frequency. The second one is interaction torque $T_i(t)$, which represents the difference between the elbow joint torque of human $T_h(t)$ and the DC motor torque $T_m(t)$, that is $T_i(t) = T_h(t) - T_m(t)$. From the definition, the value of cross-correlation between the DC motor input signal and the interaction torque are calculated as follows:

$$\begin{aligned} R_{vi}(\tau) &= E[V(t)T_i(t+\tau)] \\ &= E[V(t)(T_h(t+\tau) - T_m(t+\tau))] \end{aligned} \quad (1)$$

Here, it should be noted that the DC motor input, human muscle torque, and DC motor torque commonly originate from SEMG signal $S(t)$, and those are can be

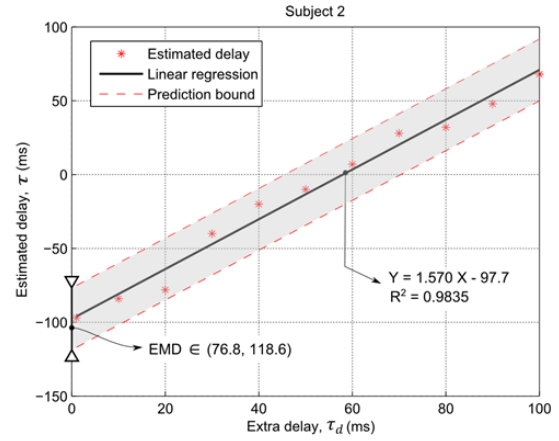


Fig. 10 Estimation using a cross correlation analysis

expressed as follows:

$$\begin{aligned} V(t) &= k_v S(t - (\tau_d + \tau_f)) \\ T_h(t) &= k_h S(t - \tau_h) \\ T_m(t) &= k_m S(t - (\tau_d + \tau_f + \tau_m)) \end{aligned} \quad (2)$$

where τ_h , τ_d , τ_f , and τ_m denote the EMD of human muscle, artificial time delay, low-pass filter phase delay, and the motor response time delay, respectively. Therefore, (1) can be rewritten as follows:

$$\begin{aligned} R_{mi}(\tau) &= k_v k_h E[S(t - \tau_d - \tau_f)S(t - \tau_h + \tau)] \\ &\quad - k_v k_m E[S(t - \tau_d - \tau_f)S(t - \tau_d - \tau_f - \tau_m + \tau)] \end{aligned} \quad (3)$$

Here we can assume that torque gain of human muscle k_h is sufficiently much larger than DC motor gain k_m . Then, (3) is approximately rewritten as follows:

$$\begin{aligned} R_{mi}(\tau) &= k_v k_h E[S(t - \tau_d - \tau_f)S(t - \tau_h + \tau)] \\ &= k_v k_h R_{ss}(\tau + \tau_d + \tau_f - \tau_h) \end{aligned} \quad (4)$$

where $R_{ss}(\tau)$ is auto-correlation of SEMG signal. Because auto-correlation has maximum value at time delay 0, cross-correlation $R_{mi}(\tau)$ should have maximum value at $\tau = \tau_h - \tau_f - \tau_d$. This means that maximum value of cross-correlation $R_{mi}(\tau)$ exists when time delay of assistive device (that is, $\tau_d + \tau_f$) is exactly consistent to the EMD of human muscle τ_h . From this relation, we investigated time lag at which cross-correlation $R_{mi}(\tau)$ has maximum value. In Fig. 10, the horizontal axis represents the artificial time delay τ_d which is control variable during the experiments, and the vertical axis represents the time lag $\tau = \tau_h - \tau_d - \tau_f$ which is obtained by the cross-correlation analysis between $V(t)$ and $T_i(t)$. Therefore, y-axis intercept implies time lag between the EMD of human muscle and the filter delay (that is, $\tau = \tau_h - \tau_f$), and x-axis intercept implies the artificial delay which made no time lag between DC motor torque and muscle torque (that is, $\tau = \tau_h - \tau_d - \tau_f = 0$). In other words, $\tau = 0$ means that total time delay $\tau_d + \tau_f$ generated by assistive device consistent with the EMD.

From Fig. 10, EMD of subject 2, τ_h is estimated as $76.8 \sim 118.6(ms)$, and the optimal artificial delay τ_{d*} , which does not make time lag or lead between DC motor torque and Human muscle torque, is estimated as $40 \sim 60(ms)$ using the zero crossing of the prediction bounds. In this range of the artificial delay τ_{d*} , we expected that human can control assistive device as own muscle because they have the same time delay from onset time of SEMG. This investigation show the timing of the assistive torque can affect the performance of the movement and may stimulate a great interest among mechatronic engineers who work on the physical human robot interaction.

5. CONCLUSION

We focused on the timing of the assistive force of the SEMG-driven assistive device. The inevitable delay was examined as about 40(ms), which comes from the signal processing (about 30(ms)) and the motor response time (about 10(ms)). First, we measured the EMD of each subject (see Table 1). Then, the movement task was carried out varying the artificial delay up to 100(ms). In the 1(ms) case, the timing of the assistive force precedes the intended force from the muscle due to EMD, while the assistive force lag behind the user intended force as increasing the artificial delay (see Fig. 10). The mean of the performance time of each condition was compared, and the minimum performance time was observed in the range from 20(ms) to 50(ms). The mean of the performance time in 1(ms) case was significantly larger than that of $20 \sim 60(ms)$ cases. The cross correlation analysis also showed the optimal artificial delay, which indicates the consistence of the timing of EMD and the timing of assistive force from the device. This investigation show the timing of the assistive torque can affect the performance of the movement and may stimulate a great interest among mechatronic engineers who study on the physical-human-robot interaction field.

ACKNOWLEDGEMENT

This work was supported by the R&D program of MKE/KEIT [1003521, ADL Support System for the Elderly and Disabled]

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